KS3 DIGITAL TECHNOLOGY

APP DEVELOPMENT WITH USER STORIES

How real apps get made: the types of app and the roles in an app team, writing user stories with acceptance criteria and prioritising them, wireframing screens and flows, building with event-driven programming and data, testing with a proper test plan, and launching, measuring and improving an app ethically using agile sprints.

MR BUTLER'S DIGITAL TECHNOLOGY

LESSONS

1 How apps are made
2 User stories and priorities
3 Wireframes and flows
4 Building with events and data
5 Testing and bugs
6 Launch, measure, improve

LESSON 1 OF 6

HOW APPS ARE MADE

Apps can be native (built for one platform), web apps (run in a browser) or cross-platform.

STARTER · 5 MINUTES

DO NOW

Name an app you use every day. How many people do you think it took to make it?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe types of app, the roles in an app development team and the stages of app development

SUCCESS CRITERIA: I CAN...

- ✓ I can compare native, web and cross-platform apps
- ✓ I can describe roles in an app development team
- ✓ I can describe the agile approach to app development

TYPES OF APP

- Native app: built for one platform (iOS or Android), downloaded from an app store
- Web app: runs in a browser; works on any device
- Cross-platform: one codebase that runs on both iOS and Android
- Each has trade-offs in speed, cost and features

KEY WORDS

NATIVE APP	An app built for one specific platform, such as iOS or Android.
WEB APP	An app that runs in a web browser.
CROSS-PLATFORM	Built once to run on more than one platform.
AGILE	An approach to development that builds and improves software in short cycles with frequent feedback.
SPRINT	A short, fixed period in which an agile team delivers a working improvement.

THE APP TEAM

- Product manager: decides what to build and why
- UX/UI designer: designs how it works and looks
- Developers: write the code (front end and back end)
- Testers: find bugs; data analysts: measure how it's used

AGILE DEVELOPMENT

- Build in short cycles called sprints (often 1-2 weeks)
- Each sprint delivers a small working improvement
- Get feedback from users, then adapt
- The opposite of planning everything up front ("waterfall")

ACTIVITY · 20 MINUTES · PAIRS

APP AUTOPSY

1. Choose an app you know well (e.g. a messaging, music or school app).
2. Decide whether it's native, web or cross-platform (or find out).
3. List the team roles that would have been needed to build it.
4. Describe one improvement you think was delivered in a recent update, and how an agile team might have decided on it.

In your workbook, answer:

? Which roles would be needed for your app, and what would each do?

? How might an agile team have decided on a recent update?

Your workbook: Lesson 1 · tick each step off as you go

QUICK CHECK

1. What is a native app?

A An app built for one specific platform
B An app made in a jungle
C A website
D A game console

2. Which type of app runs in a web browser?

A A native app
B A desktop app only
C A web app
D A console game

3. What is a sprint in agile?

A A running race
B A short cycle that delivers a small working improvement
C A type of app
D A bug

4. Who decides what an app team builds next and why?

A The tester only
B The product manager
C The app store
D The user's parents

QUICK CHECK: ANSWERS

1. **A: An app built for one specific platform**
   e.g. iOS only.

2. **C: A web app**
   Works on any device.

3. **B: A short cycle that delivers a small working improvement**
   Often 1-2 weeks.

4. **B: The product manager**
   Priorities and purpose.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Compare native and web apps.

Look back at today's success criteria: can you tick them all?

- ✓ I can compare native, web and cross-platform apps
- ✓ I can describe roles in an app development team
- ✓ I can describe the agile approach to app development

LESSON 2 OF 6

USER STORIES AND PRIORITIES

User stories describe what a user needs and why: "As a ..., I want ..., so that ...".

STARTER · 5 MINUTES

DO NOW

You're making an app for the school canteen. What do pupils, canteen staff and parents each want from it?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Write user stories with acceptance criteria and prioritise them using MoSCoW and a product backlog

SUCCESS CRITERIA: I CAN...

- ✓ I can write user stories in the correct format
- ✓ I can write clear acceptance criteria
- ✓ I can prioritise user stories using MoSCoW

USER STORIES

- As a [type of user], I want [goal], so that [reason]
- Focus on the user's need, not the technology
- Different users have different stories
- e.g. "As a pupil, I want to see today's menu, so that I can decide whether to bring lunch"

KEY WORDS

USER STORY	A short description of a feature from a user's view: "As a ..., I want ..., so that ...".
ACCEPTANCE CRITERIA	Specific, testable conditions that show a user story is complete.
PRODUCT BACKLOG	A prioritised list of all the user stories and work for a product.
MOSCOW	A way to prioritise: Must have, Should have, Could have, Won't have (this time).
MINIMUM VIABLE PRODUCT	The simplest version of a product that is still useful to real users.

ACCEPTANCE CRITERIA

- Conditions that must be true for the story to be "done"
- Specific and testable
- Often written as Given ... When ... Then ...
- e.g. Given it's a school day, when I open the app, then I see today's menu with prices

PRIORITISE

- Product backlog: the list of all stories, in priority order
- MoSCoW: Must have, Should have, Could have, Won't have (this time)
- MVP: the minimum viable product, the smallest version that's useful
- Build the Musts first, release, learn, then add more

ACTIVITY · 30 MINUTES · GROUPS OF 3

STORY WORKSHOP

1. Choose an app idea: a homework planner, a school club finder, or a local bus times app for Fermanagh.
2. Identify three types of user and write at least six user stories.
3. Write acceptance criteria (Given … When … Then …) for two of your stories.
4. Prioritise all stories with MoSCoW and circle your MVP.

In your workbook, answer:

? Write two user stories for different users.

? Write acceptance criteria for one story.

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. What is the correct format for a user story?

A As a ..., I want ..., so that ...
B Once upon a time ...
C Dear user ...
D If ... then ... else

2. What are acceptance criteria?

A The app store rating
B A list of users
C The app's price
D Testable conditions that show a story is done

3. What does the M in MoSCoW stand for?

A Money
B Must have
C Maybe
D More

4. What is an MVP?

A A marketing video
B The final version
C The best player
D The simplest version of an app that's still useful

QUICK CHECK: ANSWERS

1. **A: As a ..., I want ..., so that ...**
   Who, what, why.

2. **D: Testable conditions that show a story is done**
   Define done.

3. **B: Must have**
   Essential features.

4. **D: The simplest version of an app that's still useful**
   Minimum viable product.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Write a user story for a pupil using a school sports club app.

Look back at today's success criteria: can you tick them all?

- ✓ I can write user stories in the correct format
- ✓ I can write clear acceptance criteria
- ✓ I can prioritise user stories using MoSCoW

LESSON 3 OF 6

WIREFRAMES AND FLOWS

Before coding, designers sketch screens as wireframes, simple layouts showing where content and buttons go.

STARTER · 5 MINUTES

DO NOW

Draw your phone's home screen from memory in 60 seconds. What did you include?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Design app screens and navigation using wireframes, user flows and common mobile UI patterns

SUCCESS CRITERIA: I CAN...

- ✓ I can create a low-fidelity wireframe
- ✓ I can draw a user flow for a task
- ✓ I can apply common UI patterns and accessibility guidelines

WIREFRAMES

- A simple sketch of a screen's layout
- Boxes for images, lines for text, labelled buttons
- Low-fidelity: quick pencil sketches; high-fidelity: detailed mock-ups
- Focus on layout and function, not colours

KEY WORDS

WIREFRAME	A simple sketch of a screen's layout.
USER FLOW	A diagram of the steps a user takes to complete a task.
UI PATTERN	A common design solution users already recognise, such as a tab bar.
MOBILE-FIRST	Designing for small screens first, then adapting for larger ones.
TOUCH TARGET	The area of the screen that responds to a tap.

USER FLOWS

- The steps a user takes to complete a task
- Start → screens and decisions → goal reached
- Fewer steps usually means a better experience
- Each user story becomes a flow

PATTERNS AND ACCESSIBILITY

- Familiar patterns: bottom tab bar, search bar, cards, back button
- Mobile-first: design for small screens, then scale up
- Touch targets big enough for fingers
- Good contrast, readable text, labels for screen readers

ACTIVITY · 30 MINUTES · PAIRS

WIREFRAME SPRINT

1. Using your app idea from last lesson, choose your three Must-have user stories.

2. Draw a user flow for one of them.

3. Sketch low-fidelity wireframes for each screen in that flow (use the phone templates).

4. Swap with another pair and test: can they complete the task by "tapping" your paper screens? Note any confusion.

In your workbook, answer:

? Draw or describe your user flow.

? What confused your testers, and what will you change?

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What is a wireframe?

A A simple sketch of a screen's layout
B A type of cable
C A wire fence
D A finished app

2. What does a user flow show?

A The speed of the internet
B The number of users
C The app's code
D The steps a user takes to complete a task

3. Which is a common mobile UI pattern?

A Random layouts on every screen
B A button smaller than a pinhead
C A bottom tab bar
D A hidden menu with no icon

4. What does mobile-first mean?

A Designing for small screens first
B Building phones first
C Designing for TVs first
D Ignoring phones

QUICK CHECK: ANSWERS

1. **A: A simple sketch of a screen's layout**
   Layout before colour.

2. **D: The steps a user takes to complete a task**
   Task journey.

3. **C: A bottom tab bar**
   Familiar to users.

4. **A: Designing for small screens first**
   Then scale up.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why designers create wireframes before coding.

Look back at today's success criteria: can you tick them all?

- ✓ I can create a low-fidelity wireframe
- ✓ I can draw a user flow for a task
- ✓ I can apply common UI patterns and accessibility guidelines

LESSON 4 OF 6

BUILDING WITH EVENTS AND DATA

Apps are event-driven: code runs when events happen, such as a button tap, a timer or a new message.

STARTER · 5 MINUTES

DO NOW

When you tap "Send" in a messaging app, what do you think the code has to do?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain event-driven programming and how apps use variables, lists, stored data and APIs

SUCCESS CRITERIA: I CAN...

- ✓ I can explain event-driven programming
- ✓ I can explain how apps store data locally and in the cloud
- ✓ I can explain how apps use APIs

EVENT-DRIVEN

- An event is something that happens: a tap, a swipe, a timer, a message arriving
- An event handler is the code that runs when that event happens
- "When Save button clicked → do this"
- The app waits for events, then responds

KEY WORDS

EVENT	Something that happens that a program can respond to, such as a tap or timer.
EVENT HANDLER	Code that runs when a particular event happens.
VARIABLE	A named store for a value that can change while a program runs.
DATABASE	An organised store of data, often on a server.
API	Application programming interface: rules that let programs request data or services from each other.

DATA IN APPS

- Variables: store single values (score, username)
- Lists: store many values (homework items)
- Local storage: on the device; works offline
- Cloud database: shared, synced across devices

APIS

- Apps request data or services from other systems
- Weather, maps, payments, login with Google
- The app sends a request; the API sends back data (often JSON)
- Saves building everything yourself

ACTIVITY · 25 MINUTES · PAIRS

EVENT MAPPING

1. For your app, list five events users or the system could trigger (e.g. tap Save, swipe to delete, 8am timer).

2. For each event, write what the event handler should do in pseudocode, including input checks.

3. Decide what data your app stores: which variables, which lists, local or cloud, and why.

4. Identify one API your app could use. If you have time, build one screen in App Inventor or Thunkable.

In your workbook, answer:

? Write the event handler for one event.

? Should your app store data locally or in the cloud? Explain.

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What is an event?

A Something that happens, like a tap or timer
B A variable
C A party
D A database

2. What is an event handler?

A Code that runs when an event happens
B A wireframe
C A type of storage
D A person who organises parties

3. Why do apps save data to storage?

A Variables are too fast
B Data in variables is lost when the app closes
C They don't
D Storage is free

4. Which storage lets data sync across a user's devices?

A The screen
B A cloud database
C Local storage only
D A variable

QUICK CHECK: ANSWERS

1. **A: Something that happens, like a tap or timer**
   Triggers code.

2. **A: Code that runs when an event happens**
   Responds to events.

3. **B: Data in variables is lost when the app closes**
   Persistence.

4. **B: A cloud database**
   Shared and synced.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain what is meant by event-driven programming.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain event-driven programming
- ✓ I can explain how apps store data locally and in the cloud
- ✓ I can explain how apps use APIs

LESSON 5 OF 6

TESTING AND BUGS

Testing checks that an app works as intended and meets its acceptance criteria.

STARTER · 5 MINUTES

DO NOW

What's the most annoying bug you've found in an app or game? How do you think it got through?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Plan and carry out testing using a test plan with normal, boundary and erroneous data, and write useful bug reports

SUCCESS CRITERIA: I CAN...

- ✓ I can write a test plan with expected results
- ✓ I can choose normal, boundary and erroneous test data
- ✓ I can write a clear bug report

WHY TEST?

- Bugs annoy users, lose data and can create security holes
- Testing checks the app meets its acceptance criteria
- Test as you go, not just at the end
- Test on different devices, screen sizes and connections

KEY WORDS

TEST PLAN	A list of tests, test data and expected results.
NORMAL DATA	Sensible data that a program should accept.
BOUNDARY DATA	Data at the edges of the allowed range.
ERRONEOUS DATA	Data that a program should reject.
BUG REPORT	A record describing a problem so developers can find and fix it.

TEST DATA

- Normal: sensible data that should work (age 13)
- Boundary: at the edges of what's allowed (age 11 and 18)
- Erroneous: data that should be rejected (age "cat" or -5)
- Each test has an expected result

BUG REPORTS AND BETA TESTING

- Bug report: title, steps to reproduce, expected vs actual, device, screenshot
- Severity: how bad is it?
- Beta testing: real users try the app before the full release
- AI-written code needs testing too

ACTIVITY · 25 MINUTES · PAIRS

BREAK IT!

1. Your teacher shares a simple (deliberately buggy) app or web form, e.g. a club sign-up form.
2. Write a test plan with at least eight tests, including normal, boundary and erroneous data.
3. Run the tests and record the actual results.
4. Write two bug reports for the most serious bugs you found.

In your workbook, answer:

? Give one boundary test and one erroneous test from your plan.

? Write one of your bug reports.

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. What is boundary data?

A Random data
B Data that's always wrong
C Data at the edges of what's allowed
D Data from another app

2. What is erroneous data?

A Data at the limit
B Data that should be rejected
C Encrypted data
D Normal sensible data

3. Which part of a bug report helps developers recreate the problem?

A The tester's favourite colour
B The app's logo
C The price
D Steps to reproduce

4. What is beta testing?

A The final release
B Testing only by developers
C Testing the letter beta
D Real users trying an almost-finished app before launch

QUICK CHECK: ANSWERS

1. **C: Data at the edges of what's allowed**
   e.g. min and max values.

2. **B: Data that should be rejected**
   Wrong type or out of range.

3. **D: Steps to reproduce**
   Exact steps.

4. **D: Real users trying an almost-finished app before launch**
   Real-world feedback.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

An app accepts a quantity from 1 to 10. Give one normal, one boundary and one erroneous test value.

Look back at today's success criteria: can you tick them all?

- ✓ I can write a test plan with expected results
- ✓ I can choose normal, boundary and erroneous test data
- ✓ I can write a clear bug report

LESSON 6 OF 6

LAUNCH, MEASURE, IMPROVE

Launching an app involves app store review, a privacy policy and clear descriptions.

STARTER · 5 MINUTES

DO NOW

Have you ever felt tricked into tapping something in an app? What happened?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how apps are launched, measured and improved, and evaluate ethical issues including privacy and dark patterns

SUCCESS CRITERIA: I CAN...

- ✓ I can describe how apps are published and maintained
- ✓ I can explain how analytics and feedback drive improvements
- ✓ I can identify dark patterns and design ethical alternatives

LAUNCH

- App stores review apps for safety, privacy and quality
- A privacy policy must explain what data is collected
- Clear description, screenshots and age rating
- Plan for updates: bugs, new OS versions, security

KEY WORDS

APP STORE REVIEW	Checks an app store carries out before publishing an app.
ANALYTICS	Data showing how people use an app or website.
A/B TESTING	Comparing two versions of something with different groups of users.
DARK PATTERN	A design trick that pushes users into doing something they didn't intend.
PRIVACY POLICY	A document explaining what data an app collects and how it's used.

MEASURE AND IMPROVE

- Analytics: how many users, which features, where they quit
- Feedback: reviews, surveys, support messages
- A/B testing: show two versions to different users and compare
- Each sprint: pick the most valuable improvements from the backlog

ETHICAL APPS

- Business models: paid, subscription, ads, in-app purchases
- Dark patterns trick users: hidden costs, hard-to-cancel subscriptions, fake countdowns
- Ethical design: honest choices, easy to cancel, privacy by default
- Children's apps must follow the UK Children's Code

ACTIVITY · 30 MINUTES · GROUPS OF 3

LAUNCH PLAN

1. For your app, write the app store description, age rating and three privacy label points.
2. Choose a business model and explain why it's fair to your users.
3. Choose three analytics measures you'll track and one A/B test you'll run.
4. Check your design for dark patterns using the table, and fix any you find.

In your workbook, answer:

? Which business model did you choose, and why is it fair?

? Describe one A/B test you'll run.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What does app store review check?

A Nothing
B Only the price
C Safety, privacy, content and quality
D Only the logo

2. What is A/B testing?

A Testing two phones
B Testing the alphabet
C Comparing two versions with different groups of users
D A spelling test

3. What is a dark pattern?

A Night mode
B A bug
C A dark colour scheme
D A design trick that pushes users into doing something they didn't intend

4. What does the UK Children's Code require?

A Dark mode
B Free apps only
C High privacy settings by default for children's apps
D No apps for children

QUICK CHECK: ANSWERS

1. **C: Safety, privacy, content and quality**
Before publishing.

2. **C: Comparing two versions with different groups of users**
Evidence-based design.

3. **D: A design trick that pushes users into doing something they didn't intend**
Manipulative design.

4. **C: High privacy settings by default for children's apps**
Protects children's data.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how analytics help an app team decide what to improve.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe how apps are published and maintained
- ✓ I can explain how analytics and feedback drive improvements
- ✓ I can identify dark patterns and design ethical alternatives

END OF TOPIC

APP DEVELOPMENT WITH USER STORIES

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.